

Data-Driven Approaches to Optimizing Urban Mobility in New York City: A Thematic Literature Review

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1 Introduction

New York City (NYC) faces persistent urban mobility challenges, including uneven public transit coverage, road congestion, and highly variable travel demand influenced by factors like weather and time of day. In recent years, researchers and planners have increasingly turned to data-driven approaches to address these issues.

This literature review synthesizes recent studies that leverage data analytics and machine learning to optimize NYC’s transportation system in three key areas: (1) improving public transportation coverage and accessibility, (2) implementing and evaluating congestion pricing, and (3) forecasting travel demand with attention to temporal and weather-related patterns. The goal is to connect these findings to NYC’s transportation landscape and identify how data-driven insights can inform policy.

2 Public Transportation Coverage and Accessibility in NYC

Equitable coverage of public transportation is a cornerstone of sustainable urban mobility. Data-driven studies in NYC have focused on identifying “transit deserts” and understanding how gaps in transit service correlate with travel behavior. For example, one study combined taxi GPS records with transit infrastructure data to measure and map transit accessibility (e.g., the time required to reach a transit stop from a given location) [1].

The analysis revealed that transit accessibility, along with demographic and socioeconomic variables, significantly influences taxi trip demand across NYC. In fact, transit accessibility, population density, education levels, income, and local employment were identified as the strongest predictors of taxi usage in each neighborhood [1].

These results suggest that areas with poor transit service see higher reliance on taxis and ride-hailing, highlighting inequities in coverage. Ensuring that under-served communities have adequate transit options is also a priority in NYC’s policy frameworks. New York State’s Climate Leadership and Community Protection Act (CLCPA) mandates the identification of “Disadvantaged Communities” to ensure that frontline and otherwise under-served populations benefit from cleaner transportation and infrastructure investments [2]. In line

with this, analyses can overlay transit coverage maps with socioeconomic indicators to pinpoint neighborhoods most in need of improved service. By observing the following map of NYC with hotspots identified, we can see areas most in need of additional transport.

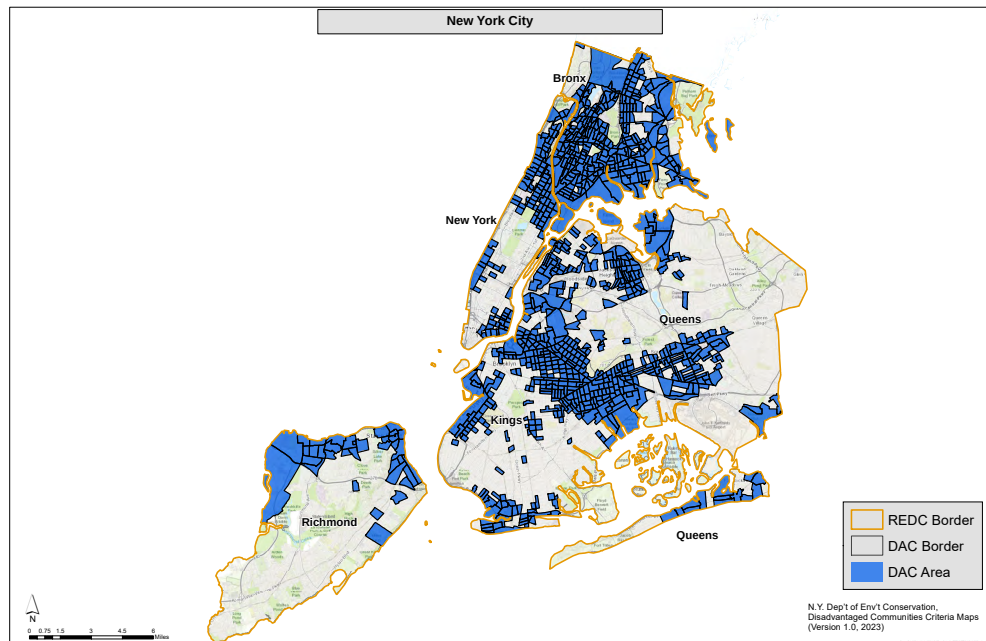


Figure 1: Heatmap of NYC taxi demand in underserved areas.[\[2\]](#)

Private and public initiatives have started addressing such gaps. For instance, the Clean Transit Access Program (CTAP) by Dollaride uses data to target NYC’s transit-poor zones where commuter “dollar vans” operate, and is facilitating the transition of these services to electric vehicles [\[3\]](#). By providing fleet owners serving disadvantaged areas with electric vans and charging infrastructure, this program aims to reduce emissions and improve service reliability in communities that historically lack robust public transit [\[3\]](#). These results allow us to narrow our focus in terms of location and target, while expanding our options.

Beyond identifying gaps, data-driven models can guide how to fill them. Clustering techniques are used to group similar neighborhoods or travel patterns, which helps transit planners design routes that serve clusters of unmet demand. Classification models can predict which areas or population groups are likely to be transit-under-served based on features like distance to the nearest subway or bus line, population density, and car ownership rates. Such analyses directly inform initiatives like bus network redesigns and microtransit pilots.

Overall, the literature emphasizes that integrating diverse data sources (transit schedules, ridership data, taxi trips, census data) provides a richer picture of mobility needs. This multi-source approach enables planners to optimize public transportation coverage in NYC by prioritizing new routes or services for the areas that need them most, thereby advancing accessibility [\[1\]](#).

3 Time and Weather-Related Demand Forecasting

Another area addresses how data-driven methods can forecast urban mobility demand under weather conditions. Travel demand in NYC is significantly modulated by weather events (rain, snow, extreme temperatures). In our research questions, we asked how the weather might influence public transit. This question has been studied extensively, and accurate forecasting currently enables transit agencies and mobility providers to proactively adjust service. A variety of time-series modeling techniques, from classical statistical models to modern machine learning, have been applied to NYC transportation data to capture these patterns. One prominent approach is spatio-temporal time series modeling. [Safikani et al.](#) introduced a generalized Space-Time Autoregressive (STAR) model to predict yellow taxi demand across NYC’s neighborhoods, incorporating both spatial and temporal dependencies. The STAR model exhibited superior forecasting accuracy compared to standard time-series baselines; in out-of-sample tests it outperformed traditional vector autoregression (VAR) models in predicting taxi pickups, demonstrating the value of capturing location-based correlations in demand. Notably, this framework is interpretable for operators and computationally efficient for real-time use, showing that advanced yet practical forecasting tools can be derived from data. Other studies have utilized machine learning regression algorithms (such as random forests and gradient boosting) on NYC taxi and ride-hail data. These models similarly report mean absolute percentage errors (MAPE) in the low teens, improving on simpler moving-average forecasts and thereby enabling better allocation of drivers or vehicles to meet anticipated demand spikes. This extensive research shows NYC has incredibly robust weather prediction and service adjustment. Thus we decided to ignore weather conditions in future analysis.

4 Options for expansion

Using analysis from the New York State Climate Action Council, [\[2\]](#) we can observe areas of NYC that are currently considered to be "Transit deserts". An obvious question to ask is "What is the difficulty in serving these areas to begin with?". By using the Bronx, a well known transit desert [\[9\]](#), as a case study, we can isolate our research. Plans already exist to expand the existing subway system, and have existed for some time. Yet the political will has not manifested these results. We must examine these political realities and combine them with our analysis, as they say Politics is the art of the possible, and impossible plans are of no help to those living in underserved areas. In our analysis, we find 2 major reasons the Bronx has historically been underserved. 1) The Bronx is poor. While obvious, it bears repeating. The citizens of the Bronx lack the political or economic power to make their needs known. [\[9\]](#) 2) Less pessimistically, it has been shown that Manhattan and other well served areas have a tight concentration of jobs, and thus, a tight concentration of transit hotspots. The Bronx is far more evenly distributed. This makes permanent transit fixtures less attractive.

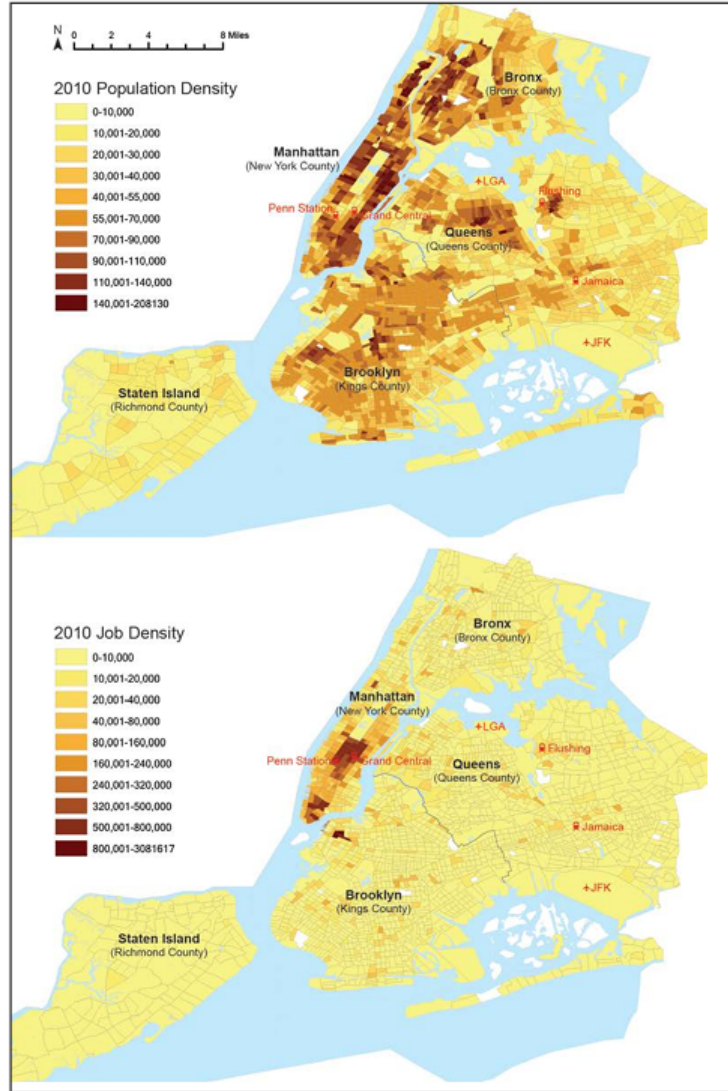


Figure 5. 2010 Population and Job Density (per square mile)

Figure 2: Heatmap of NYC job locations.[\[1\]](#)

Due to this analysis, we intend to focus our efforts on bus transportation and "ride sharing" services like Dollaride. Gonzalez et al, has already performed extensive modeling using taxi data gathered over the course of 10 months and over 147 million rides. Their methods include linear regression with examination of related coefficients. Step-wise selection, wherein the best model was chosen by seeking the model with the lowest value of Akaike Information Criterion (AIC) and smaller Residual Sum of Squares (RSS). As well as best subsets prediction to determine which variables to include in the model. Our goal is slightly different, but the methods used will be similar. [\[1\]](#)

5 Conclusion and Future Directions

New York City’s transportation system faces persistent equity and coverage issues, particularly in transit deserts like the Bronx. Existing research has identified these underserved areas using diverse data sources, revealing clear spatial disparities in public transit accessibility. Studies have shown that low-income, transit-poor neighborhoods rely heavily on taxis and ride-hailing services, underscoring the need for better public infrastructure in these regions.

In response, some efforts—such as the CTAP and microtransit initiatives—have targeted these gaps with flexible, data-informed solutions. However, permanent infrastructure like subways remains politically and logistically difficult to expand, especially in low-density or economically marginalized areas. As a result, scalable, adaptive solutions like enhanced bus service have emerged as the most viable option.

Building on this foundation, our work will focus on using historical taxi data to identify patterns of unmet transit demand across NYC, with a focus on transit deserts. By analyzing where people are consistently paying for taxis in areas with poor public transit, we aim to pinpoint locations that would most benefit from expanded coverage.

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